

STATS 503, Homework 8

1. [8 POINTS] ISLP chapter 9 exercise #2
2. [8 POINTS] ISLP chapter 9 exercise #3
3. [4 POINTS] For this question, use graph paper, a drawing application, or Python.

(a) Suppose

$$f(X_1, X_2) = \frac{1}{\sqrt{2}}X_1 - \frac{1}{\sqrt{2}}X_2 - 1.$$

Draw the hyperplane defined by $f(X_1, X_2) = 0$. Indicate the region of possible inputs x_1, x_2 for which $f(X_1, X_2) > 0$ with a “+” sign. Indicate the region of possible inputs x_1, x_2 for which $f(X_1, X_2) < 0$ with a “−” sign.

(b) Suppose an SVM classifier is fitted to data, resulting in the following decision rule:

$$\hat{y}(x_1, x_2) = \begin{cases} + & \text{if } f(x_1, x_2) > 0 \\ - & \text{otherwise.} \end{cases}$$

Suppose further that this SVM is associated with a margin of $m = \sqrt{2}$.¹ Draw the margin lines.

- (c) What class label (+ or −, as defined above) does this SVM predict for the following points: (1,4); (1,1); (2, -5); (2, -1); (4,2)?
 - (d) Suppose that these five points were part of the training data, and their true labels, given in the same order, are −, −, +, +, −. Calculate the corresponding slack values (ξ_i ’s) for each of the five points.
4. This question uses the `crabs.csv` data, with the five body measurements as predictors and `species` as the response. Omit all other variables including `sex`. We will assess error in terms of misclassification rate.
- (a) [2 POINTS] Fit a linear SVM to the data for a range of values of the C hyperparameter, to predict `species` from the five numerical measurements. Plot the cross-validated estimate of the error as a function of C .
 - (b) [4 POINTS] Now, fit nonlinear SVMs with polynomial kernels. Consider at least three values of `degree`. For each, plot the cross-validated estimate of the error as a function of C .
 - (c) [4 POINTS] Finally, fit nonlinear SVMs with radial kernels. Consider at least three values of `gamma`. For each, plot the cross-validated estimate of the error as a function of C .

¹Under some conventions for how SVMs are written, this corresponds to a decision function of the form

$$\tilde{f}(X_1, X_2) = \frac{1}{2}X_1 - \frac{1}{2}X_2 - \frac{1}{\sqrt{2}},$$

in which $\|\beta\| = 1/\sqrt{2}$. There are unfortunately many notations used for SVMs!